ML Based Automatic Modulation Classification

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INTRODUCTION

- **Automatic Modulation Classification (AMC)** identifies the **modulation type** of received signals **without requiring prior knowledge of transmitter parameters**
- It functions as an **important intermediate step** between **signal detection** and the **demodulation process** in **wireless communication systems**
- The project utilizes the **RadioML dataset**, which contains **over 2.5 million signal samples** across **24 different modulation schemes**
- Signals in the dataset are represented as **raw In-phase (I) and Quadrature (Q) components**, along with corresponding **Signal-to-Noise Ratio (SNR) labels**
- Both **traditional machine learning models (KNN, Random Forest)** and **deep learning models (CNN)** are used and compared, especially for performance under **noisy conditions**

OBJECTIVES

- Understand **Automatic Modulation Classification (AMC)** as a core component of modern digital communication receivers
- Use a **labeled RadioML dataset** to perform supervised modulation classification under varying SNR conditions
- Analyze dataset characteristics, including **modulation types, features, and noise levels**
- Reduce dataset size (**32,000 samples**) to improve computational efficiency and prevent overfitting
- Develop and evaluate a **machine learning-based AMC system** using appropriate feature representations
- Assess model performance using **metrics like accuracy, loss, and confusion matrix** across different SNR values
- Address limitations of traditional ML and extend to **deep learning models with hyperparameter tuning** for improved performance

MOTIVATION

- Rapid growth of wireless communication systems demands **efficient and intelligent signal analysis techniques**
- Traditional modulation recognition methods rely on **manual feature extraction**, which is time-consuming and less adaptive
- Performance of conventional approaches **degrades significantly in noisy (low SNR) environments**
- Increasing need for **automated and real-time modulation classification** in applications like cognitive radio and spectrum monitoring
- Availability of large datasets (e.g., RadioML) enables **data-driven approaches using ML and Deep Learning**
- Deep learning models can **automatically learn complex signal patterns**, reducing dependency on handcrafted features
- Aim to improve **accuracy, robustness, and scalability** of modulation classification for next-generation wireless systems

PROBLEM STATEMENT

Accurately identifying modulation schemes from received signals without prior knowledge of transmitter parameters remains a significant challenge in modern wireless communication systems. Traditional signal processing approaches rely heavily on handcrafted feature extraction, which often fails to generalize well under varying and noisy channel conditions. In particular, performance tends to degrade at low Signal-to-Noise Ratio (SNR) levels. With the increasing complexity and demand of wireless technologies, there is a need for robust, scalable, and data-driven modulation classification methods that can automatically learn meaningful features directly from raw signal data.

METHODOLOGY

- A total of **32,000 signal samples** were selected to train the model
- Focused on **8 modulation schemes** (4ASK, BPSK, QPSK, 16PSK, 16QAM, FM, AM-DSB-WC, 32APSK) for easier analysis and visualization
- Selected a specific **SNR value of -10 dB** (from the range -20 to 20) to prepare **X_data and Y_data**
- Applied **one-hot encoding** to modulation labels and later converted them into **integer format** for model training
- Split the dataset into **80% training and 20% testing**
- Implemented **Machine Learning models (KNN, Random Forest)** with and without hyperparameter tuning
- Developed a **Deep Learning model** to classify modulation schemes at the selected SNR level

BLOCK DIAGRAM

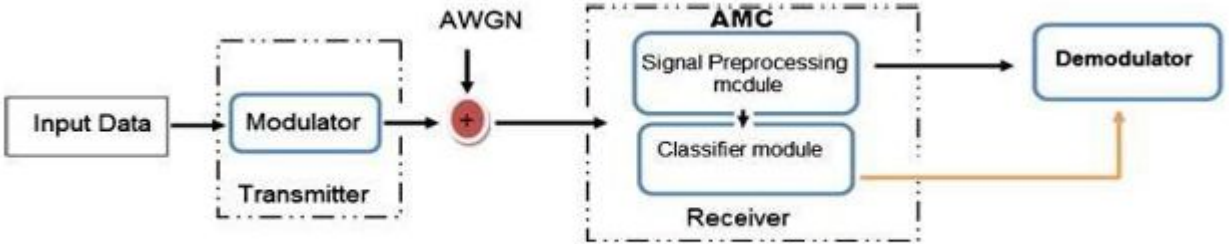

Fig. 1. Block Diagram for Automatic Modulation Classification

BLOCK DIAGRAM DESCRIPTION

- **Signal Generation / Dataset**
  Modulated signals such as **BPSK, QPSK, QAM** are generated or obtained from datasets like **RadioML2018**
- **Channel Model**
  Signals pass through wireless channel effects including **AWGN noise, Rayleigh fading, and phase offset**, simulating real-world conditions
- **I/Q Sample Extraction**
  The receiver captures **complex In-phase (I) and Quadrature (Q) samples**, representing the baseband signal
- **Preprocessing**
  Data is **normalized and reshaped** into suitable formats (tensors) for input into machine learning and deep learning models
- **Feature Extraction (CNN Layers)**
  Convolutional Neural Networks (CNNs) automatically extract important patterns and features from raw signal data
- **Classification Layer**
  A **fully connected layer** processes extracted features to predict the modulation class
- **Output**
  The system outputs the **predicted modulation type**

BLOCK DIAGRAM-DEEP LEARNING MODEL

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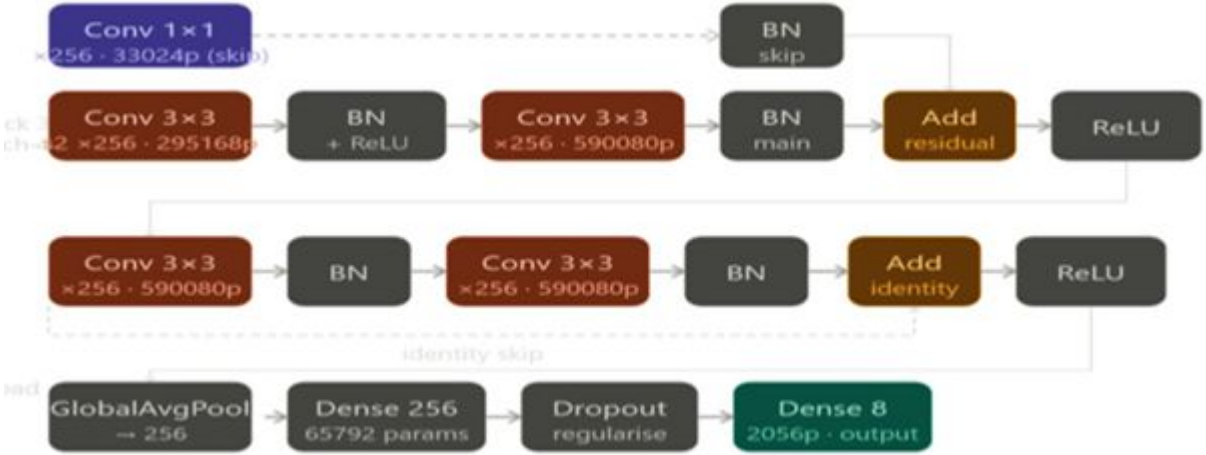


DEEP LEARNING MODEL DESCRIPTION

- **Input Representation**
  Input consists of **complex baseband I/Q samples** represented as two channels: **In-phase (I)** and **Quadrature (Q)**, arranged as a **2D tensor** suitable for convolution
- **Initial Convolution Layer**
  A **3×3 convolution layer** extracts low-level features like **amplitude variations and phase information**, followed by **Batch Normalization (BN)** and **ReLU activation**
- **Deep Feature Extraction**
  Multiple **convolutional blocks (Conv → BN → ReLU)** are stacked to learn hierarchical features such as **symbol transitions, constellation patterns, and modulation-specific signatures**
- **Residual Learning (Skip Connections)**
  Uses **ResNet-style connections**, including **identity skip** and **projection (1×1 convolution)** to improve gradient flow and training stability
- **Downsampling & Channel Expansion**
  Uses **stride (s=2)** for downsampling and increases filters (**64 → 128 → 256**), enabling **reduced spatial size and richer feature representation**

DEEP LEARNING MODEL DESCRIPTION

- **Global Feature Aggregation**
  Applies **Global Average Pooling** to convert feature maps into a compact vector, reducing **model complexity and overfitting**
- **Fully Connected Layers**
  Features are passed through **Dense layers with Dropout**, enabling the model to learn final classification boundaries
- **Output Layer**
  Uses **Softmax activation** to generate a **probability distribution over modulation classes**, selecting the highest as output
- **Model Outcome**
  The CNN learns features **directly from raw I/Q data**, achieving **higher accuracy and better performance in noisy conditions** compared to traditional ML methods

PROGRESS AND RESULTS

K-Nearest Neighbors (KNN)

- KNN classifies signals based on the **majority class of nearest neighbors**
- **Steps involved:**
  - Compute **Euclidean distance** between signals
  - Select **k nearest neighbors**
  - Assign the **most frequent modulation class**
- The trained model predicts modulation types on test data
- Achieved a test accuracy of approximately **56%**

PROGRESS AND RESULTS

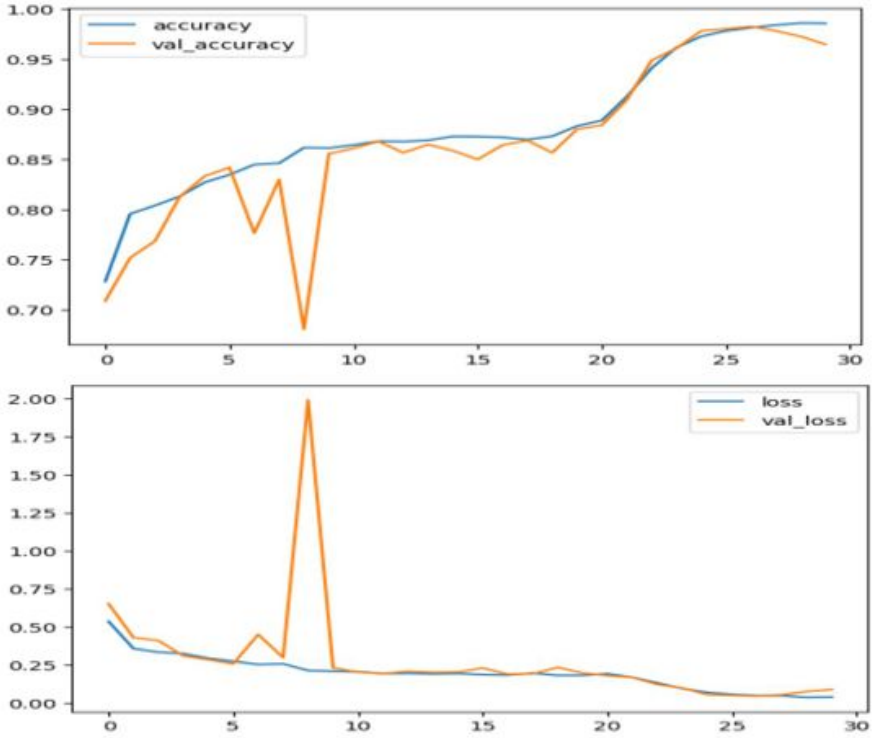
Random Forest (with Hyperparameter Tuning)

- Random Forest is an **ensemble learning method** using multiple **decision trees**
- Learns relationships between **signal features and modulation classes**
- **Key Parameters:**
  - **n_estimators:** Number of trees (best: 200)
  - **max_depth:** Tree depth (best: None for deeper trees)
  - **min_samples_leaf:** Prevents overfitting (best: 4)
  - **n_jobs = -1:** Utilizes all CPU cores
- Final prediction is made using **majority voting** of all trees
- Achieved improved test accuracy of approximately **66%**, outperforming KNN

PROGRESS AND RESULTS

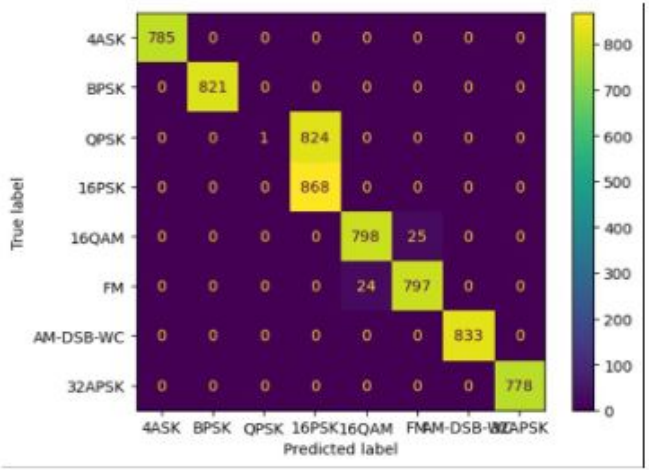
- A **CNN-based model** was implemented to classify modulation signals at **SNR = -10 dB**
- The model consists of **2,812,360 parameters (~10.73 MB)**, with **2,807,880 trainable parameters** enabling effective learning of complex signal patterns
- A small number of **non-trainable parameters (4,480)** arise from **Batch Normalization**, which helps stabilize training
- Initially, the model achieved around **85% accuracy with 10 epochs**, capturing basic modulation features
- Increasing training to **30 epochs** improved feature learning and weight optimization
- Final model achieved approximately **98% classification accuracy**, showing strong performance even under noisy conditions

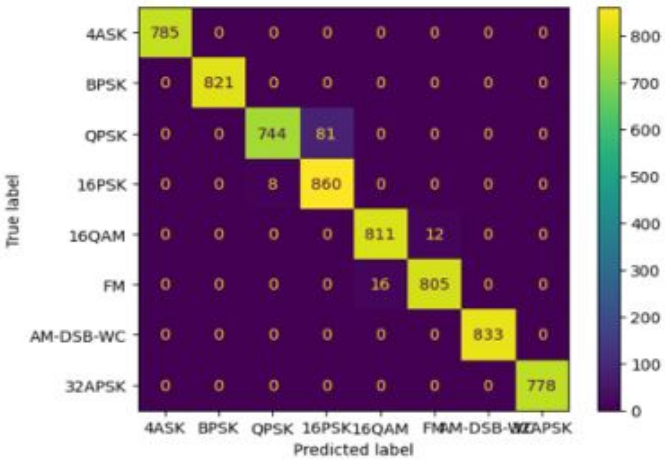
CNN Model Training and Validation Accuracy and Loss vs. Epochs



Confusion Matrix for the Deep learning model

- Rows represent **actual classes**, columns represent **predicted classes**
- Diagonal values indicate **correct classifications**
- Off-diagonal values indicate **misclassifications**
- First one represents **85% accuracy** model's confusion matrix and Second one of **98% accuracy**





RESULTS ANALYSIS

- Accuracy improvement from **~85% to ~98%** is reflected in the confusion matrices
- At **85% accuracy**, noticeable **misclassifications** occur between similar modulation schemes (e.g., **QPSK vs 16PSK**, **16QAM vs FM**)
- Indicates the model had not fully learned **distinctive features** at lower epochs
- After **30 epochs**, the confusion matrix becomes **almost perfectly diagonal**
- **Off-diagonal values are significantly reduced**, showing fewer errors
- Model learns more **discriminative features**, improving class separation
- Results in **near-perfect classification accuracy** with minimal confusion between modulation types

CONCLUSION

- Initial performance using **ML models (KNN, Random Forest)** achieved around **66% accuracy**
- Implementation of a **CNN on raw I/Q data** improved accuracy significantly to **~98% (after 30 epochs)**
- **Misclassifications reduced**, as observed in the **confusion matrix**
- CNN effectively learns **complex signal features** and distinguishes between **similar modulation schemes**
- Results in **more robust and reliable classification performance**

FUTURE WORK

- Extend the model to handle **multiple SNR levels** instead of a single SNR
- Incorporate **SNR-based subclassification** for better generalization
- Improve performance under **real-world noisy conditions**

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- X. Y. Zeng, A. B. Clark, and M. Q. Smith, “Deep learning-based modulation classification using CNNs,” in Proc. IEEE Global Communications Conference, Singapore, 2023, pp. 123–12.